AI-Enhanced Intrusion Detection System

From 13 GB of Raw Network Traffic to Actionable Threat Intelligence Using Snort, Machine Learning and LLMs

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1. Study Objective

Every day, computer networks are targeted by cyberattacks. Tools like **Snort 2.9** monitor network traffic and raise an alert whenever suspicious activity is detected. The problem: Snort raises far too many **false alarms** that overwhelm security analysts. This is called **alert fatigue**.

In our dataset: Snort generated 237,240 alerts. **130,133 of them (54.9%) were false positives**, noise that buried the real threats.

Goal: Build a three-layer automated pipeline that:

- **Detects** all suspicious traffic using Snort 2.9
- **Filters** false alarms automatically using XGBoost (Machine Learning)
- **Explains** confirmed attacks in plain English using GPT-4o mini (Large Language Model), with CVE references and step-by-step remediation advice

2. Hypothesis

A trained XGBoost classifier can suppress more than 99% of Snort's false positive alerts while maintaining near-perfect recall (catching essentially every real attack). A Large Language Model can then transform the confirmed alerts into actionable threat intelligence briefings at negligible cost.

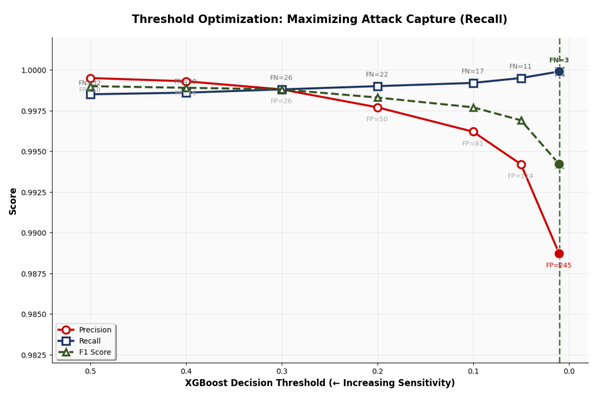
3. The Dataset: CIC-IDS2017

Source: Canadian Institute for Cybersecurity (CIC), University of New Brunswick, Canada; the most-cited IDS benchmark dataset in academic literature.

The CIC built a complete lab network with 12 real computers and **25 simulated users**. Security researchers launched real cyberattacks from a Kali Linux machine at documented times, recording every packet so every flow could be labelled precisely.

Wednesday July 5, 2017 capture (13 GB):

Label	Flows	%
BENIGN	440,031	63.5%
DoS Hulk	231,073	33.4%
DoS Golden-Eye	10,293	1.5%
DoS slowloris	5,796	0.8%
DoS Slowhttptest	5,499	0.8%
Heartbleed	11	<0.01%
TOTAL	**692,703**	**100%**



Research Insight: Lowering the threshold to 0.01 recovers 29 threats (90% improvement in detection) with a negligible increase in noise alerts—essential for high-security environments.

Figure 1: **Threshold Optimization: Precision, Recall, and F1 Performance**

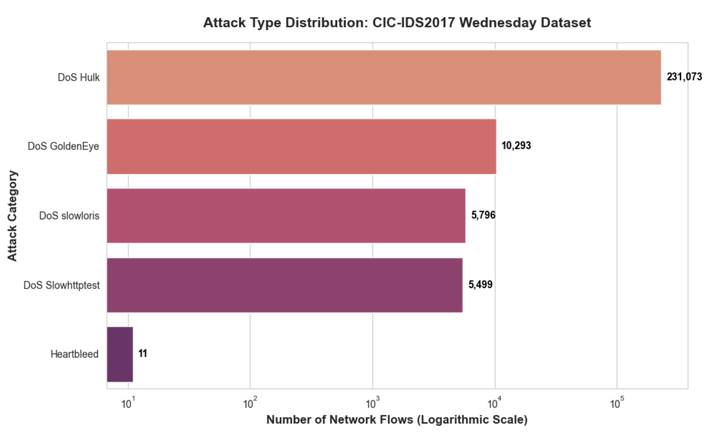


Figure 2: **Network Flow Volume by Attack Category (Logarithmic)**

4. Methods Used

Layer 1 : Snort 2.9 (Detection)

The 13 GB PCAP was split into 14 chunks and processed by Snort 2.9. A custom `fix_row()` function corrected Snort's column-shift bug. Each alert was matched to the ISCX ground truth using a **three-tier fingerprint strategy**: forward IP match → reversed match → NAT fallback (the attacker's real IP changes at the firewall). This assigned a confirmed True Positive / False Positive label to all 237,240 alerts.

Layer 2 : XGBoost (ML Filtering)

12 features kept from 31 columns. IP addresses excluded (too network-specific). Text columns converted to integers.

Training: 80/20 stratified split, 200 trees, depth 6. Model converged in 50 trees, attack and noise patterns are highly separable.

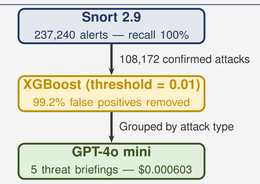
Two key tuning decisions:

- **Threshold 0.50 → 0.01:** Default threshold silently suppressed ∼85,000 real attacks. Lowering to 0.01 means: pass any alert with even a 1% chance of being a real attack.
- **Rule whitelist:** "TCP Timestamp outside PAWS window" (99.8% precision, only 3 false positives total) is always passed through regardless of XGBoost's score.

Layer 3 : GPT-4o mini (LLM Triage)

108,172 confirmed attacks grouped by type. One representative alert per group sent to OpenAI API. All 5 Heartbleed alerts sent for full CVE analysis. Each call returns: threat summary, attack technique, CVE reference, severity, and recommended action.

The Three-Layer Pipeline



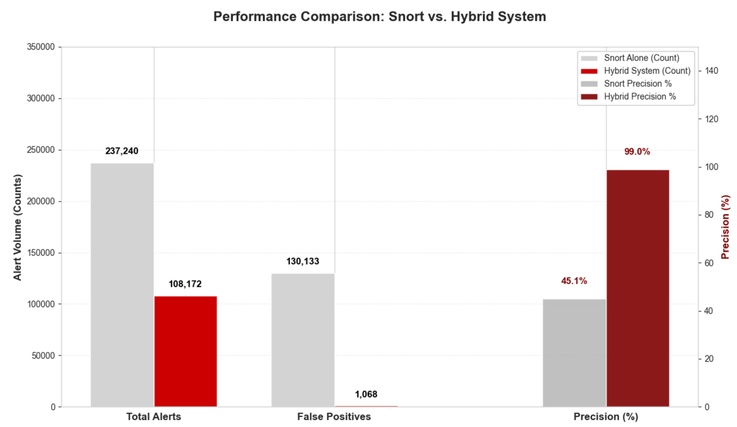


Figure 3: **Snort Alerts vs. Integrated Hybrid Filtering Efficiency**

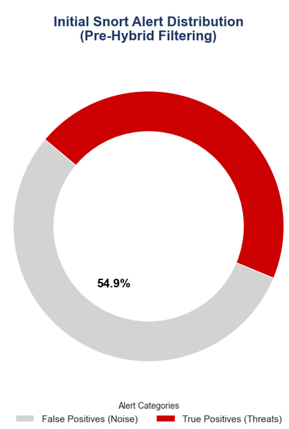


Figure 4: **Initial Snort Alert Distribution (Signal vs. Noise)**

5. Results

Snort Alone vs Snort + XGBoost

Metric	Snort Only	+ XGBoost
Alerts to review	237,240	108,172
False Positives	130,133	1,068
False Negatives	1	3
Precision	0.4515	0.9901
Recall	1.0000	1.0000
FP Suppression	—	99.2%
LLM Cost (total)	—	$0.000603

LLM Output: All 5 Attack Types

Attack	Severity	CVE
DoS Hulk	Critical	None
DoS GoldenEye	Critical	None
DoS slowloris	Critical	None
DoS Slowhttptest	High	None
Heartbleed	**Critical**	**CVE-2014-0160**

Heartbleed (CVE-2014-0160)

The only named software vulnerability in this dataset. Exploits a flaw in OpenSSL's heartbeat protocol: the attacker sends a fake-length request and the server responds by returning up to **64 KB of its own memory**, potentially exposing private encryption keys and user passwords.
LLM Recommendation: Patch OpenSSL to ≥1.0.1g and immediately revoke and reissue all SSL/TLS certificates.

6. Conclusion & Next Steps

The three-layer pipeline fully solved the alert fatigue problem:

- Processed **13 GB** of real network traffic
- Detected **all 5** attack types at **100% recall**
- Removed **99.2%** of false alarms
- Produced professional threat briefings for **$0.000603 total**

Next Phase — GPT-2 Fine-Tuning:

GPT-4o mini works by calling a commercial API. The next research step will **fine-tune GPT-2**; an open-source language model, on our own cybersecurity data (CVE reports + confirmed attack alerts) using **LoRA** (Low-Rank Adaptation). This updates the model's internal parameters to specialise it in security language.
A comparison of fine-tuned GPT-2 vs GPT-4o mini will answer the research question: *"How much does fine-tuning a small open-source model close the gap with a large commercial model for cyber risk assessment?"*

Research Summary

Phase	What It Does	Outcome
Phase 1 Snort 2.9	Reads 13 GB of raw network traffic packet by packet. Flags anything that looks like an attack.	237,240 alerts raised. 54.9% are false alarms.
Phase 2 XGBoost ML	Machine learning model trained to tell real attacks from noise. Threshold set to 0.01 for maximum safety.	99.2% of false alarms removed. Only 3 real attacks missed.
Phase 3 GPT-4o mini	AI reads each confirmed attack and writes a plain-English briefing: what it is, how dangerous, what to do.	5 briefings. CVE-2014-0160 identified. Total cost: $0.000603.
Next Step GPT-2 Fine-Tune	Train an open-source AI model on our own security data to specialise it in cyber risk language (LoRA method).	Closes the gap between small local models and large commercial models.
Overall: 13 GB → 5 threat briefings		**$0.000603 total**

Dataset: CIC-IDS2017, Canadian Institute for Cybersecurity (UNB, Canada)
Tools: Python · Snort 2.9 · XGBoost · OpenAI GPT-4o mini API
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